

ASSIGNMENT #2

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: Stephen Gibbs

Due Date: Wednesday February 05 by 5:00pm in my office

Number of points.....100

Your score..... 41

Note: All problems are worth 100 points. Your final score will be the average value

To convert 1000MMSCFD to mass flow rate:

$$n = \frac{V \cdot P}{R \cdot T}, \text{ where } T \text{ and } P \text{ are temperature and pressure under standard conditions. } \checkmark$$

$$n = \frac{1 \times 10^9 \frac{\text{ft}^3}{\text{day}} \cdot 14.696 \text{ psia}}{10.7239 \frac{\text{psia} \cdot \text{ft}^3}{\text{lbmol} \cdot \text{R}} \cdot 519.67 \text{ R}} = 2637052 \frac{\text{lbmol}}{\text{day}} = 1196146 \frac{\text{kmol}}{\text{day}}$$

$$1196146 \frac{\text{kmol}}{\text{day}} \cdot 18.015 \frac{\text{kg}}{\text{kmol}} = 21548570 \frac{\text{kg}}{\text{day}}$$

After converting to $\frac{\text{kg}}{\text{s}}$, we get $250 \frac{\text{kg}}{\text{s}}$.

The subscripts used for P, T, H, S, etc, correspond to the stream numbers from the diagram.

Stream 1:

Starting with $P_1 = 8000 \text{ kPa}$ and $T_1 = 480 \text{ }^\circ\text{C}$, from the steam tables:

$$H_1 = 3347.35 \frac{\text{kJ}}{\text{kg}} \text{ and } S_1 = 6.6573 \frac{\text{kJ}}{\text{kg} \cdot \text{ }^\circ\text{C}}.$$

Stream 2:

After pass through the high-pressure turbine, $P_2 = 700 \text{ kPa}$ and using $T_c @ 700 \text{ kPa}$, $T_2 = 164.956 \text{ }^\circ\text{C}$. We can then use those values and the steam tables to get the 2-phase H and S values:

$$H_{2V} = 2760.13 \frac{\text{kJ}}{\text{kg}}, H_{2L} = 696.60 \frac{\text{kJ}}{\text{kg}}, S_{2V} = 6.7007 \frac{\text{kJ}}{\text{kg} \cdot \text{ }^\circ\text{C}} \text{ and } S_{2L} = 1.9905 \frac{\text{kJ}}{\text{kg} \cdot \text{ }^\circ\text{C}}.$$

Because the turbine process is reversible: $S_1 = S_2 = 6.6573 \frac{\text{kJ}}{\text{kg} \cdot \text{ }^\circ\text{C}}$

$$\text{Using the lever-rule to get the composition of the stream: } X_2 = \frac{S_2 - S_{2L}}{S_{2V} - S_{2L}} = \left(\frac{6.6573 - 1.9905}{6.7007 - 1.9905} \right) \frac{\text{kJ}}{\text{kg} \cdot \text{ }^\circ\text{C}} = 0.99$$

We can then use the composition to calculate the theoretical value for H_2 :

$$H_2(\text{theo}) = H_{2V} \cdot X_2 + (1 - X_2) \cdot H_{2L} = 2739.49 \frac{\text{kJ}}{\text{kg}} \checkmark$$

We then use $H_2(\text{theo})$ to calculate the theoretical power produced by the high-pressure turbine:

$$H_2(\text{theo}) - H_1 = W_{\text{HPT}}(\text{theo}) = (2739.49 - 3347.35) \frac{\text{kJ}}{\text{kg}} = -607.86 \frac{\text{kJ}}{\text{kg}}$$

$$W_{\text{HPT}}(\text{actual}) = W_{\text{HPT}}(\text{theo}) * \text{efficiency} = -607.86 \frac{\text{kJ}}{\text{kg}} * 0.88 = -534.92 \frac{\text{kJ}}{\text{kg}}$$

$W_{\text{HPT}}(\text{actual})$ and H_1 are then used to calculate the actual value of H_2 :

$$H_2 = W_{\text{HPT}}(\text{actual}) + H_1 = -534.92 \frac{\text{kJ}}{\text{kg}} + 3347.35 \frac{\text{kJ}}{\text{kg}} = 2812.43 \frac{\text{kJ}}{\text{kg}} \quad \checkmark$$

Stream 3:

$$P_3 = 700 \text{ kPa}$$

$$T_3 = 164.956^\circ\text{C} \quad \text{need to recalculate } T$$

$$S_3 = S_2 = 6.6573 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}} \quad \text{need to recalculate } S \quad \left. \begin{array}{l} T \\ S \end{array} \right\} (-5)$$

$$H_3 = H_2 = 2812.43 \frac{\text{kJ}}{\text{kg}}$$

Stream 4:

After passing through the second turbine, LPT, $P_4 = 8 \text{ kPa}$ and using $T_c @ 8 \text{ kPa}$, $T_4 = 41.53^\circ\text{C}$, we can follow the same steps used in Stream 2 to calculate the rest of the values for Stream 4:

$$S_{4L} = 0.5922 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}, S_{4V} = 8.2241 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}, S_4 = S_3 = 6.6573 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}$$

$$H_{4L} = 173.75 \frac{\text{kJ}}{\text{kg}}, H_{4V} = 2575.39 \frac{\text{kJ}}{\text{kg}}, H_4 = 2159.9 \frac{\text{kJ}}{\text{kg}} \quad \text{should use } S_3 \text{ real}$$

$$W_{\text{LPT}}(\text{actual}) = -652.53 \quad \text{and } H_4 \text{ real} = ? \quad (-5)$$

Stream 5:

After passing through the condenser, $P_5 = 8 \text{ kPa}$.

$$\text{Since } X_5 = 0, H_5 = H_{4L} = 173.75 \frac{\text{kJ}}{\text{kg}} \text{ and } S_5 = S_{4L} = 0.5922 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}$$

$$Q_c = H_5 - H_4 = 173.75 \frac{\text{kJ}}{\text{kg}} - 2159.9 \frac{\text{kJ}}{\text{kg}} = -1986.15 \frac{\text{kJ}}{\text{kg}}$$

need to use H_4 real

Stream 6:

After passing through the first pump, LPP:

$$P_6 = 700 \text{ kPa and } S_6 = S_5 = 0.5922 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}$$

ideal

Using trial and error we can look for what temperature at 700 kPa results in $0.5922 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}$.

$$T_6 = 41.56 ^\circ\text{C}$$

And we once again use the steam tables to find $H_6 = 174.47 \frac{\text{kJ}}{\text{kg}}$. *should calculate H_6 real (-2)*

Work required for the first pump:

$$W_{\text{LPP}} = H_6 - H_5 = 174.47 \frac{\text{kJ}}{\text{kg}} - 173.75 \frac{\text{kJ}}{\text{kg}} = 0.720 \frac{\text{kJ}}{\text{kg}}$$

Stream 7:

Using flowrates of *why? (-2)* 125 kg/s for the incoming flows we can calculate the temperature for stream 7:

2. *(-2)* $m_{\text{bypass}} * (T_{\text{bypass}} - T_7) = m_6 * (T_7 - T_6)$, plugging in the known values and solving for T_7 , $T_7 = 61.698$.

$P_7 = 700 \text{ kPa}$ and using the steam tables, $H_7 = 258.60$ and $S_7 = 0.8515$. $\times$

Stream 8:

After passing through the second pump, HPP:

$$P_8 = 8000 \text{ kPa and } S_8 = S_7 = 0.8513 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}$$

ideal

should calculate H_8 real (-2)

Using trial and error we can find that $T_8 = 61.698$ and using the steam tables we find that $H_8 = 266.35 \frac{\text{kJ}}{\text{kg}}$

$$W_{\text{HPP}} = H_8 - H_7 = 266.35 \frac{\text{kJ}}{\text{kg}} - 258.60 \frac{\text{kJ}}{\text{kg}} = 7.75 \frac{\text{kJ}}{\text{kg}}$$

$$Q_b = H_1 - H_8 = 3347.35 \frac{\text{kJ}}{\text{kg}} - 266.35 \frac{\text{kJ}}{\text{kg}} = 3081 \frac{\text{kJ}}{\text{kg}}$$

Power:

by what flow rate? (-2)
Multiplying each value of work or heat gives us the power generated or consumed by each process:

$$P_{\text{Qb}} = 770 \text{ MW} \quad P_{\text{Qc}} = -248 \text{ MW} \quad P_{\text{HPT}} = -133 \text{ MW}$$

$$P_{\text{LPT}} = -81 \text{ MW} \quad P_{\text{LPP}} = 0.09 \text{ MW} \quad P_{\text{HPP}} = 1.9 \text{ KW}$$

$$P_{\text{Gen}} = 463 \text{ MW}$$

$$P_{\text{Cons}} = 772 \text{ MW}$$

1) Maximum flow rate for the bypass is 500 MMSCFD $\times (-5)$

2) Power generated = 463 MW $\times (-2)$

3) Efficiency = $P_{\text{Gen}} / P_{\text{Cons}} = 463 \text{ MW} / 772 \text{ MW} \times 100\% = \underline{59.9\%}$
 (-2)

4) Condenser water flow rate:

$-P_{\text{Qc}} = Q$ absorbed by feed water

$$Q = C \cdot m \cdot dT \Rightarrow \frac{Q}{C \cdot dT} = m \Rightarrow \frac{248000 \frac{\text{kJ}}{\text{s}}}{4.184 \frac{\text{kJ}}{\text{kg} \cdot \text{C}} \cdot 11^\circ \text{C}} = 5388 \frac{\text{kg}}{\text{s}},$$
$$5388 \frac{\text{kg}}{\text{s}} \cdot \frac{\text{m}^3}{1000 \text{ kg}} \cdot \frac{264.172 \text{ gal}}{\text{m}^3} \cdot \frac{60 \text{ s}}{\text{min}} = \underline{27.3 \text{ GPM}}$$

reference? (-2)

5) NHV Coal = 26 mil btu/ton = 7.54 MW/ton $\times (-2)$

$$P_{\text{Qb}} = 133 \text{ MW} \Rightarrow 133 / 7.54 = 17.6 \text{ ton/day}$$

$\swarrow$ conversion? (-1)

Question 6?

(-25)